

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2014

CL314 Water Resources Engineering and Management

Date: 15th May, 2014

Time: 10:00 a.m. To 01:00 p.m.

Day: Thursday

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1(a) Discuss, with a neat sketch, different zones of storage in a reservoir. Also, explain the significance of valley storage. [05]
- (b) Define: Safe yield, Secondary yield [02]
- Q - 2(a) Explain in detail, different types of investigations necessary for the reservoir planning. [05]
- (b) The mean monthly flow in a river during a year is observed as follows: [09]

Sr. No.	Month	Flow (cumecs)	Sr. No.	Month	Flow (cumecs)
1	January	50	7	July	52
2	February	40	8	August	80
3	March	32	9	September	110
4	April	24	10	October	85
5	May	15	11	November	70
6	June	24	12	December	60

Using mass-curve analysis, obtain the maximum uniform demand that can be assured by storage of 2800 cumec-day. Also, locate the periods of excess flow and depletion during a year, if reservoir is half-full at the beginning of the year. Assume each month of 30 days.

OR

- Q - 2(a) What is a mass curve of inflow? How is it used to determine the storage capacity of a reservoir for a constant demand? [07]
- (b) What are the detrimental effects of sedimentation? Suggest suitable remedial measures required to be adopted to control reservoir sedimentation. [07]

- Q -3(a) Discuss different reservoir losses and measures to reduce them. [04]
- (b) What are the different methods of flood control? Explain, clearly, the non-structural methods of flood damage control. [06]
- (c) Explain: (i) Penstock (ii) Turbine [04]

OR

- Q - 3(a) Discuss causes and effects flood. [02]
- (b) Route the flood through a channel, using Muskingum method, for the inflow hydrograph given as: [08]

Sr. No.	Date & Time	Inflow (m^3/s)	Sr. No.	Date & Time	Inflow (m^3/s)
1	1-08-95, 6.00 a.m.	10	5	2-08-95, 6.00a.m.	40
2	Noon	30	6	Noon	32
3	6.00 p.m.	70	7	6.00p.m.	20
4	Mid-night	50	8	Mid-night	10

Take storage constant and dimensionless constant for the given channel reach may be taken as 0.5 day and 0.12 respectively. Calculate attenuation and time of travel.

- (c) Explain 'Surge tank' and its importance in a hydroelectric project. [04]

SECTION - II

- Q - 4 How bed formation occurs due to sediment transport in channels? [07]
- Q - 5(a) What are the needs of Water Resources Planning? [05]
- (b) Which economic considerations will be influenced in the planning of a water resources project? [04]
- (c) How drought can be tackled by water management? [05]

OR

- Q - 5(a) List out various steps involved in the planning of water resources project. [05]
- (b) For better planning and management of project which strategies are carried out? [04]
- (c) Discuss drought mitigation options. [05]

Q - 6(a) Design guide bunds (or banks) and launching apron required to be provided for a [09]
bridge across a river whose total waterway is 655 m. The design flood discharge is
13000 m³/s, which may be increased by 15% for design of launching apron. Mean size
of the river bed material is 0.3 mm. Also draw the neat sketch showing cross section of
guide bank.

(b) Write the requirements of good lining material.

[05]

Useful formulae:

$$C_o = \frac{-KX + 0.5\Delta t}{K - KX + 0.5\Delta t}; \quad C_1 = \frac{KX + 0.5\Delta t}{K - KX + 0.5\Delta t}; \quad C_2 = 1 - C_o - C_1$$

$$T = 0.04 \text{ to } 0.06 Q^{1/3}$$

$$R = 0.48 \left(\frac{Q}{f} \right)^{1/3} \quad f = 1.76 \sqrt{d}$$

$$d_{\min} = 0.023 \text{ to } 0.046 (U_{\max})^2$$

$$Q_2 = C_o I_2 + C_1 I_1 + C_2 Q_1$$